

Predicting Risk of Poverty in California

Fong, Marcela, Aziza

1. Problem Statement and Research Question

Poverty is one of California's most persistent problems, affecting millions of people and driving inequality across health, education, and economic outcomes. Existing social programs generally react after the fact rather than identifying who is at risk beforehand. This project uses machine learning to predict individual poverty risk using data from the American Community Survey (ACS) 2024 5-Year PUMS for California.

Our primary research question is: Can demographic characteristics, employment conditions, education level, and health insurance coverage predict whether an individual in California falls below the poverty threshold? Two supporting questions examine how education, employment, hours worked, and wages influence poverty likelihood, and how age, marital status, citizenship, and insurance coverage create differences in risk across the population.

We used *poverty_status*, a binary variable derived from the ACS income-to-poverty ratio, as the target variable. We started with the full California ACS file (285 variables), sampled 25,000 records using DuckDB, and selected 27 variables based on their theoretical relevance to poverty. After preprocessing, the final dataset has 23,266 records split 80/20 into 18,612 training and 4,654 test observations, with 87.7% non-poverty and 12.3% poverty, a class imbalance that influenced our modeling decisions.

2. Data Quality, Feature Selection, and Preprocessing

FER (Fertility) was removed due to high missingness. Employment-related variables (WKHP, WAGP, COW, INDP, OCCP, ESR) were filled with 0 since missing employment data typically means the individual is not working. DRIVESP and SCHL were imputed with the median, and LANX and ENG with the mode.

Feature selection was guided by economic theory and the ACS codebook, covering four areas: education (SCHL), employment and income (ESR, WKHP, WAGP, COW, INDP, OCCP), demographics (AGEP, SEX, RAC1P, MAR, NATIVITY, CIT, ENG, LANX, DIS, DRIVESP), and health insurance (HICOV, HINS1 through HINS7). WAGP and WKHP are negatively correlated with poverty status, while Medicaid shows a positive correlation as a safety-net program for lower-income populations. Multicollinearity among predictors was low, so all variables were included in the logistic regression.

3. SHAP Feature Importance Analysis

SHAP values were used to interpret feature contributions using a baseline Random Forest model, showing both the magnitude and direction of each variable's contribution to prediction outcomes. Wage income (WAGP) is the most influential predictor, confirming that income is the primary structural driver of poverty classification. Lower values consistently increase predicted poverty probability while higher values reduce it. Health insurance indicators (HINS4 and HINS1) are the next strongest group. Individuals without employer-sponsored insurance and those enrolled in Medicaid are more likely to be classified below the poverty threshold, with Medicaid functioning more as a marker of existing vulnerability than a cause. Marital status, age, hours worked, and employment status all contribute meaningfully. Education shows a moderate protective effect, though its direct contribution is smaller

than income and insurance, consistent with the interpretation that education reduces poverty risk primarily through wages and employment type rather than acting as an independent predictor. Industry, race, citizenship, sex, disability, and occupation contribute smaller but still measurable effects.

4. Model Results and Performance Comparison

Since only 12.3% of observations are in the poverty class, overall accuracy is not a useful metric. A model that always predicted non-poverty would hit 87.7% without learning anything. We focused on ROC-AUC and poverty recall as the primary evaluation criteria, since the goal is to correctly identify individuals who are actually at risk.

Metric	Logistic Regression	Random Forest	XGBoost	Best Model
Overall Accuracy	88.0%	83.6%	76.9%	LR (misleading)
ROC-AUC	0.840	0.853	0.860	XGBoost
Poverty Recall	0.17	0.62	0.83	XGBoost
Poverty Precision	0.51	0.39	0.33	Logistic Reg.
Poverty F1	0.26	0.48	0.47	Random Forest
False Negatives	472	220	100	XGBoost
False Positives	96	543	974	Logistic Reg.

4.1 Logistic Regression

WAGP has the strongest coefficient at -5.92, indicating that higher income greatly reduces the likelihood of being in poverty. However, the model correctly identifies only 100 out of 572 actual poverty cases, misclassifying 472 vulnerable individuals as non-poor. Without any correction for class imbalance, Logistic Regression defaults to the majority class, producing high accuracy but very low minority class sensitivity. Its coefficients, such as marital status (0.59), employment status (0.30), age (0.28), citizenship (0.27), Medicaid coverage (0.51), and hours worked (0.22), are easy to explain to a non-technical audience, making it the easiest to interpret of the three models despite its limited screening ability.

4.2 Random Forest

Using *class_weight='balanced_subsample'*, the Random Forest achieved 83.6% accuracy and an AUC of 0.853, correctly identifying 352 of 572 poverty cases and missing 220. The F1-score of 0.48 is the best of the three models. The improvement in recall from 0.17 to 0.62 relative to Logistic Regression is attributable to the balanced subsampling correction and the ensemble structure, which captures non-linear relationships and feature interactions that Logistic Regression cannot represent. WAGP is the strongest predictor at 24.5%, followed by HINS1 (12.7%), HINS4 (12.0%), age (7.3%), marital status (6.8%), hours worked (6.5%), and education (4.7%).

4.3 XGBoost

XGBoost achieved the highest AUC of 0.860 and poverty recall of 0.83. Trained with *scale_pos_weight*=7.13 to compensate for the class imbalance, it correctly identified 472 of 572 poverty cases with only 100 false negatives. This was the fewest false negatives of any model. The trade-off of better poverty recall is an increase in false positives and a reduction in overall accuracy. However, for a policy-oriented application where the cost of missing a vulnerable individual is high, prioritizing sensitivity over precision is an appropriate trade-off. Feature importance is led by HINS1 (18.6%), WAGP (17.1%), HINS4 (15.7%), and HICOV (8.0%), reflecting XGBoost's gain-based importance metric, which rewards splits that reduce error on the minority class.

After initial training, we optimized XGBoost using RandomizedSearchCV with 30 iterations and 5-fold cross-validation, scoring directly on poverty recall rather than overall accuracy. The tuned model improved poverty recall from 0.83 to 0.85 and reduced false negatives from 100 to 85, with ROC-AUC remaining strong at 0.859. This improvement occurred because hyperparameter optimization shifted the model toward a more sensitive decision boundary that prioritizes identifying minority class observations. The best hyperparameters included shallower trees, a lower learning rate, and moderate subsampling values, which helped reduce overfitting while improving generalization on imbalanced data.

5. Final Model Selection and Conclusion

XGBoost was selected as the final model. After hyperparameter tuning, the model achieved a poverty recall of 0.85, an ROC-AUC of 0.859, and only 85 false negatives. This means the model correctly identified 487 out of 572 individuals below the poverty threshold, outperforming Logistic Regression (missed 472 poverty cases) and Random Forest (missed 220 poverty cases). In this case's context, missing a vulnerable individual is more detrimental than incorrectly flagging someone who is not poor. These results make XGBoost the best choice out of all the models run.

Across all three models, wage income is the dominant predictor of poverty risk, followed by health insurance coverage, employment status, hours worked, marital status, and age. The consistency of these findings across all three models strengthens confidence that they reflect genuine patterns in the data rather than artifacts of any single approach. Education appears in all three models but at lower relative importance, supporting the idea that its effect on poverty runs through wages and employment rather than being a direct one.

On the first supporting question, full-time employment, higher wages, and employer-sponsored insurance are the strongest employment-related protections against poverty. Underemployment, meaning low hours even when technically employed, adds to poverty risk independently of wage income. On the second supporting question, single individuals face a higher poverty risk than married households due to income pooling effects. Citizenship status and language proficiency contribute to predicted poverty risk in ways that reflect structural barriers faced by foreign-born and non-English-speaking populations.

5.1 Limitations

- The model predicts poverty at a single point in time and does not capture near-poverty vulnerability or poverty cycling over time.
- The 25,000-row sample is a small portion of California's population and would need validation against the full ACS dataset before any real deployment.
- XGBoost's false positive rate is high at the current threshold and would need adjustment based on real-world policy cost trade-offs.

- The model does not account for regional cost-of-living differences across California counties, which meaningfully affects what income level constitutes economic hardship.

5.2 Future Opportunities

- Adding housing costs and geographic indicators could improve the usefulness of regional policy and capture local economic conditions.
- We could tune the threshold using a precision-recall curve, allowing a cutoff that matches deployment cost trade-offs rather than using the default (0.5)
- Extending to a multi-year ACS panel would allow poverty transitions to be modeled over time rather than just predicting a single snapshot.
- Applying the Streamlit-deployed model to the full California ACS file would produce a county-level poverty risk map to inform regional social services planning directly.